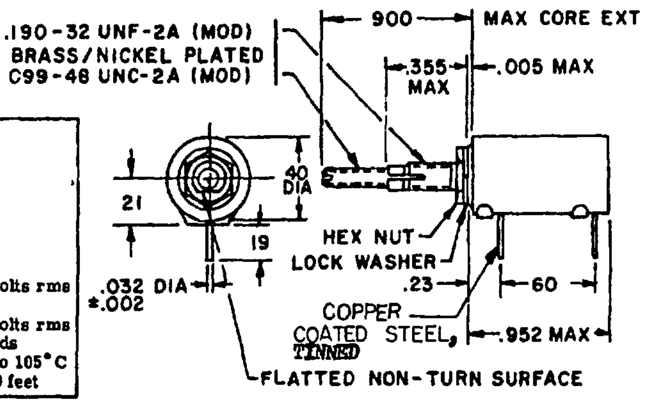
FED. SUP CLASS
5950

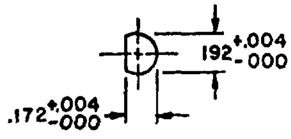
User activities
Army MI, ME, EL
Navy AS, CG, MC
Air Force 19

Reviewer activities
Army MI, MU
Navy EC, OS, SH
Air Force 11, 17, 80
DSA-EP

This military standard is approved for use by all Departments and Agencies of the Department of Defense Selection for all new engineering and design applications and for repetitive use shall be made from this document



RATINGS	
Style	LT12
Grade	3
Class	A
Maximum operating temperature	105°C
Temperature rise	15°C
Ambient temperature	90°C
Dielectric withstanding voltage (sea level)	1000 volts rms
Dielectric withstanding voltage (reduced barometric pressure)	200 volts rms
Terminal pull	5 pounds
Operating temperature	-55° to 105°C
Altitude	70, 000 feet



CHASSIS CUTOUT

INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.172	4.37	.355	9.02
.004	.10	.19	4.83	.40	10.16
.005	.13	.192	4.88	.60	15.24
.032	.18	.21	5.33	.900	22.86
.099	2.51	.23	5.84	.952	24.18

ELECTRICAL CHARACTERISTICS (INITIAL)

Dash No 1/	Type designation	Former MS part Number	Former type designation	Tuning range Inductance min	Tuning range Q min	Tuning range Test frequency	Tuning range Inductance max	Tuning range Q min	Tuning range Test frequency	Min self-resonant frequency at max specified inductance	DC resistance max Ohms	Rated DC current ma
-32	LT12V032	90546-01	LT10V094	23 µH	90	25 MHz	.40 µH	100	25 MHz	210 MHz	05	1800
-33	LT12V033	90546-02	LT10V095	29 µH	95	25 MHz	64 µH	110	25 MHz	165 MHz	053	1720
-34	LT12V034	90546-03	LT10V096	37 µH	100	25 MHz	90 µH	110	25 MHz	140 MHz	06	1650
-35	LT12V035	90546-04	LT10V097	50 µH	105	25 MHz	1 20 µH	95	7 9 MHz	125 MHz	08	1430
-36	LT12V036	90546-05	LT10V098	78 µH	105	25 MHz	1 80 µH	90	7 9 MHz	95 MHz	09	1340
-37	LT12V037	90546-06	LT10V099	1 10 µH	75	7 9 MHz	2 50 µH	65	7 9 MHz	75 MHz	10	1280
-38	LT12V038	90546-07	LT10V100	1.30 µH	65	7 9 MHz	3 60 µH	65	7 9 MHz	68 MHz	23	840
-39	LT12V039	90546-08	LT10V101	2 00 µH	70	7 9 MHz	6 00 µH	75	7 9 MHz	56 MHz	35	680
-40	LT12V040	90546-09	LT10V102	3 00 µH	80	7 9 MHz	9 00 µH	100	7 9 MHz	45 MHz	75	465
-41	LT12V041	90546-10	LT10V103	4 00 µH	80	7 9 MHz	11 00 µH	90	2 5 MHz	39 MHz	1 00	405
-42	LT12V042	90546-11	LT10V104	7 50 µH	80	7 9 MHz	18 00 µH	95	2 5 MHz	32 MHz	1.40	340
-43	LT12V043	90546-12	LT10V105	12 00 µH	70	2 5 MHz	30 00 µH	100	2 5 MHz	21 MHz	1 50	315
-44	LT12V044	90546-13	LT10V106	15 00 µH	70	2 5 MHz	44 00 µH	90	2 5 MHz	16 MHz	2 00	274
-45	LT12V045	90546-14	LT10V107	25 00 µH	75	2 5 MHz	65 00 µH	90	2 5 MHz	13 MHz	2 20	261
-46	LT12V046	90546-15	LT10V108	35 00 µH	80	2 5 MHz	90 00 µH	85	2 5 MHz	10 MHz	3 00	223
-47	LT12V047	90546-16	LT10V109	55 00 µH	80	2 5 MHz	130 00 µH	85	790 kHz	8 MHz	3 80	198
-48	LT12V048	90546-17	LT10V110	80 00 µH	70	2 5 MHz	200 00 µH	85	790 kHz	6.8 MHz	4 20	189
-49	LT12V049	90546-18	LT10V111	120 00 µH	70	790 kHz	290 00 µH	85	790 kHz	5 6 MHz	4 50	182
-50	LT12V050	90546-19	LT10V112	200 00 µH	60	790 kHz	430 00 µH	75	790 kHz	4 5 MHz	6 00	158
-51	LT12V051	90546-20	LT10V113	300 00 µH	50	790 kHz	620 00 µH	70	790 kHz	3 2 MHz	13 00	107
-52	LT12V052	90546-21	LT10V114	400 00 µH	50	790 kHz	1 00 mH	60	790 kHz	2 4 MHz	17 00	94
-53	LT12V053	90546-22	LT10V115	600 00 µH	35	790 kHz	1 70 mH	50	250 kHz	1 9 MHz	19 00	82
-54	LT12V054	90546-23	LT10V116	1 05 mH	35	250 kHz	2 70 mH	50	250 kHz	1 28 MHz	23 00	74
-55	LT12V055	90546-24	LT10V117	1 70 mH	30	250 kHz	4 00 mH	40	250 kHz	1 10 MHz	33 00	62
-56	LT12V056	90546-25	LT10V118	2 50 mH	30	250 kHz	6 00 mH	35	250 kHz	840 kHz	41 00	56
-57	LT12V057	90546-26	LT10V119	3 70 mH	30	250 kHz	9 00 mH	35	250 kHz	700 kHz	61 00	46
-58	LT12V058	90546-27	LT10V120	5 60 mH	30	250 kHz	13 00 mH	30	79 kHz	600 kHz	81 00	40
-59	LT12V059	90546-28	LT10V121	8 50 mH	30	250 kHz	22 00 mH	35	79 kHz	370 kHz	120 00	32
-60	LT12V060	90546-29	LT10V122	13 00 mH	25	79 kHz	30 00 mH	35	79 kHz	280 kHz	170 00	27
-61	LT12V061	90546-30	LT10V123	21 00 mH	20	79 kHz	47 00 mH	25	79 kHz	210 kHz	230 00	23
-62	LT12V062	90546-31	LT10V124	35 00 mH	20	79 kHz	70 00 mH	25	79 kHz	140 kHz	345 00	19

1/ The dash number added to the MS military standard number constitutes the MS part number, for example, MS 90546-32

Ⓐ Entire standard revised

APPROVED 24 March 1967 REVISED Ⓐ 27 Nov 1973

P A EL Other Cust EC 80	International interest	TITLE COILS, RADIO FREQUENCY, MOLDED VARIABLE, FERRITE CORE (PRINTED CIRCUIT MT'G) TYPES LT12V032 TO LT12V062, INCLUSIVE	MILITARY STANDARD MS 90546	
Procurement Specification MIL-C-15305		SUPERSEDES:	PAGE 1 OF 3	

FED SUP CLASS
5950

User activities. Army MI, ME, EL
Navy AS, CG, MC
Air Force 19

Reviewer activities. Army MI, MU
Navy EC, OS, SH
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DSA-EP

Electrical characteristics (Final)

Inspection Group	Allowable variation from initial measurement		Allowable % from specified minimum value in Electrical Characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification Inspection				
Group II	±2	---	--	-10
Group III	±5	±(2% + 001 ohm)	-5	-15
Group IV	±5	±(2% +.001 ohm)	-5	-10
Quality Conformance Inspection				
Group C Subgroup I	±2	---	--	-10
Subgroup II	±5	±(2% + 001 ohm)	-5	-10
Subgroup III	±5	±(2% +.001 ohm)	-5	-10

NOTES.

1. Core shall be set at the maximum specified inductance value indicated in the electrical characteristics (initial) table prior to all inspection tests. This core setting shall not be changed until electrical characteristics (Final) measurements are performed.
2. Temperature rise and terminal pull tests are applicable.
3. Barometric pressure (test condition C) is applicable.
4. Operating torque shall be a maximum of 6 ounce-inches and a minimum of 0.5 ounce-inch for tuning coil.
5. The test fixture in the diagram below shall be used for electrical measurements. The effective inductance shall be determined for all coils whenever this measurement is specified. Allowance of approximately .03 μh shall be determined for the internal inductance of the Q-meter and test fixture when measuring inductance value of 10μH or less. The shorting bar shall be made of solid copper wire (AWG #20) bent to conform to the body configuration of the coil under test (see diagram below).
6. Approximate weight:
   Dash number 32-42 ----5.8 grams
   43-52-----6.6 grams
   53-62-----7.0 grams
7. Dimensions are in inches.
8. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
9. Unless otherwise specified, tolerances are .XX±.02 and .XXX±.005.
10. For design feature purposes, this standard takes precedence over procurement document referenced herein.
11. Referenced document shall be of the issue in effect on date of invitations for bid.
12. Shock specified pulse method 213, test condition I, is applicable. Coil shall be mounted by its mounting stud and securely fastened with mounting hardware for vibration and shock testing.
13. Terminal material shall be in accordance with QQ-W-345, Type I.

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P.A EL
Other Cust EC
80

International Interest

TITLE
COILS, RADIO FREQUENCY, MOLDED VARIABLE, FERRITE CORE (PRINTED CIRCUIT MT'G) TYPES LT12V032 TO LT12V062, INCLUSIVE

MILITARY STANDARD

MS 90546

Procurement Specification
MIL-C-15305

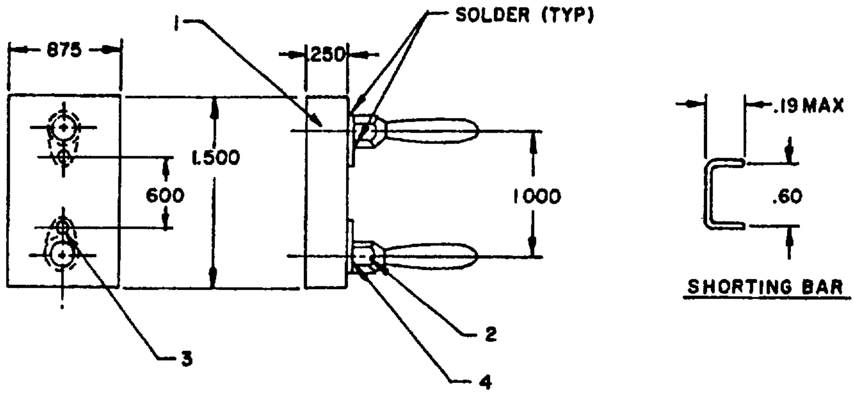
SUPERSEDES

PAGE 2 OF 3

DD FORM 1 SEP 63 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

FED SUP CLASS
5950

TEST FIXTURE FOR ELECTICAL MEASUREMENTS



Item	Description	Qty
1	Lucite block	1
2	#103 Banana plugor equivalent	2
3	Nugent PS-402-60 or equal	2
4	Copper lug	2

INCHES	MM
.19	4.83
.250	6.35
.600	15.24
.875	22.23
1.000	25.40
1.500	38.10

User activities
Army MI, ME, EL
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APPROVED 24 March 1967 REVISED Ⓐ 27 Nov 1973

P A EL Other Cust EC 80	International Interest	TITLE COILS, RADIO FREQUENCY, MOLDED VARIABLE, FERRITE CORE (PRINTED CIRCUIT MT'G) TYPES LT12V032 TO LT12V062, INCLUSIVE	MILITARY STANDARD MS 90546
Procurement Specification MIL-C-15305		SUPERSEDES:	PAGE 3 OF 3

DD FORM 1 SEP 63 672 *(Coordinated)* PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MS90546 COILS,RADIO FREQUENCY,MOLDED VARIABLE, FERRITE CORE (PRINTED CIRCUIT MT'G) TYPE LT12V032 TO LT12V062 INCLUSIVE

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 **1426** REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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